

Seth DeWhitt

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EDUCATION

Purdue University, *West Lafayette, IN* August 2023 – May 2027
B.S. in Computer Science, conc. Software Engineering & Security GPA: 3.6/4.0
Relevant Coursework: Data Structures, Analysis of Algorithms, Software Engineering, Systems Programming, Computer Architecture, Software Testing, Linear Algebra, Operating Systems, Computer Networks
Activities: Jazz Education Network (President), Purdue Jazz Band (Lead Tenor Sax.)
Awards: H.M. StarkHacks 2026 (AMD Track), Dean's List (4x), Elmhurst U. Jazz Fest Outstanding Soloist (4x)

EXPERIENCE

Software Development Engineer Intern May 2026 – August 2026
Amazon Web Services (AWS) | Erlang/OTP, JavaScript, Docker, RabbitMQ, AWS *Seattle, WA*

- Reduced RabbitMQ broker restart cycle from **2+ hours to under 2 minutes** for AmazonMQ customers via auth/authz config validation for 6 auth methods, using **Erlang/OTP** and JavaScript
- Prevented confused-deputy and race conditions by migrating AWS credential handling to thread state per-request
- Authored **412 unit, property-based, and CT integration tests**, including a decision-parity suite proving the validator's decisions match the real auth backend for identical inputs
- Contributed bugfix to upstream **RabbitMQ 4.2.9**, fixing a SigV4 signing defect that failed every S3 GET request

Research Lead & Teaching Assistant August 2025 – May 2026
The Data Mine, Purdue University | Agile, Python, TypeScript, Surreal, Firebase *West Lafayette, IN*

- Directed 11 student researchers in collaboration with the Practical AI Podcast (**1.5M+ listens**)
- Implemented RAG chatbot trained on **349+ episodes** with citation-backed responses
- Built podcast generation platform integrating LLM-produced transcripts and custom voice clone models

Software Engineer Intern June 2025 – August 2025
Chewy | Python, React, Vite, FastAPI, Docker, AWS, Terraform, Jenkins, PostgreSQL *Bellevue, WA*

- Deployed scalable & containerized development environments with custom tooling for in-house model execution on AWS ECS/Fargate, **reducing setup overhead by 80%** and supporting **15+ engineers**
- Automated distributed system maintenance with AWS Lambda and Kinesis, **lowering AWS expenses by 25%**
- Engineered **persistent storage architecture** linking Okta SSO with AWS EFS, ECS, and RDS to support secure stateful development sessions across container restarts

Undergraduate Data Science Researcher August 2023 – May 2025
The Data Mine, Purdue University | Bayer, Syngenta | Python, R, SciKit-learn, Pandas *West Lafayette, IN*

- Trained a LightGBM model on **500k+ rows** to model climate change impact on corn yields at **91% accuracy**
- Optimized model performance through feature engineering and cross-validation, reducing RMSE by **24%**
- Implemented data preprocessing and QC pipelines for **100M+ data points**, validated using mixed linear models

PROJECTS

ControVirtual | *Python, C#, Flask, PyTorch, OpenCV, LeRobot | GitHub | DevPost* April 2026

- Streamlined and upscaled use of AMD SO-101 Robot Arms for industry experts via remote use of a Meta Quest S3, voice & natural language processing, and a Vision-Language-Action (VLA) model
- Enabled spatial visibility of remote arms by livestreaming a camera feed & displaying 3D replica arms
- Honorable Mention** for "Best Use of AMD Tech" at StarkHacks 2026

SlopMop | *Python, React, Vite, Vitest, Jest, Firebase, AWS, PyTorch | GitHub* January 2026 – April 2026

- Engineering Chrome/Firefox extension that auto detects AI social media content, leading a team of 6
- Deployed a hybrid system of image detection models, maximizing accuracy & lowering response times to **<3s**

TECHNICAL SKILLS

Languages: Python, Erlang, Java, JavaScript/TypeScript, C/C++, SQL, R, C#
Frameworks & Runtimes: React, Next.js, Node.js, Flask, FastAPI, Tailwind
Databases & Tools: PostgreSQL, MongoDB, AWS, Docker, Terraform, Git, Jenkins, Pandas, Flyway, Vite